Does Daily Meditation Increase Self-Compassion? A Bayesian Multilevel Diary Study

Matthew J. Monnot

Vocentive

Author Note

Matthew J. Monnot https://orcid.org/0000-0002-9489-1777

I have no conflicts of interest to disclose. This research received no specific funding. This article reports a secondary analysis of an archival dataset (Li et al., 2021; Open Science Framework node pqf6v). The analysis protocol, the Stan model, and the analysis code were written before any outcome was modeled; these files are available in the project repository (https://github.com/mjmonnot/selfcompassion-diary-preprint). During the preparation of this work the author used large language models accessed through Cursor (Anthropic Claude, xAI Grok, OpenAI GPT, and Google Gemini) to draft and revise text, write figure code, and check reported statistics; a word processor with Clippy and autocorrect; and statistical software, so that no coefficient was obtained by hand. The author reviewed, edited, and verified the output and takes full responsibility for the content. These tools are not authors.

Correspondence concerning this article should be addressed to Matthew J. Monnot, Vocentive. Email: matt@vocentive.com

Abstract

Evaluations of brief self-compassion programs usually compare phases, leaving open whether meditation and nonmeditation days differ. With few nonmeditation days, subtracting each person's observed average can distort the separation of day-level from person-level effects. This secondary analysis used the archival 28-day diary of 65 employees in a two-week online self-compassion meditation program, with morning reports of meditation over the previous 24 hours (Li et al., 2021). Across 14 intervention days, a prespecified Bayesian two-level model treated each person's meditation propensity as latent, separating the within-person contrast between meditation and nonmeditation days from stable differences in meditation frequency. Hypotheses were higher same-day self-compassion on meditation days (primary) and lower next-day emotional exhaustion, adjusting for same-day exhaustion (secondary). Results are posterior medians, 95% credible intervals (CrIs), and directional probabilities. For same-day self-compassion, the meditation-day slope had median = 0.00 *SD*, 95% CrI [−0.09, 0.09], P($b > 0$) = .48; for next-day exhaustion, the slope had median = −0.03 *SD*, 95% CrI [−0.16, 0.11], P($b < 0$) = .65. Both CrIs spanned zero. An archival intervention diary can answer an adherence question its original phase-based analysis did not ask.

Keywords: self-compassion, meditation adherence, daily diary, latent mean centering, Bayesian multilevel models

Does Daily Meditation Increase Self-Compassion? A Bayesian Multilevel Diary Study

Self-compassion is the disposition to treat oneself with kindness when things go badly, to see one's difficulties as part of shared human experience, and to hold painful thoughts in balanced awareness (Neff, 2003). Self-compassionate people tend to rely heavily on positive cognitive restructuring when they cope with stressful events (Allen & Leary, 2010, p. 107). Self-compassion is also trainable. A structured program raised self-compassion and reduced distress in a randomized trial (Neff & Germer, 2013). Brief online formats make such training feasible for employees during a working week.

One such format was evaluated with a 28-day diary (Li et al., 2021). Sixty-five employees completed a control week, two weeks of guided online meditation (a body scan in the first week and self-compassionate breathing in the second), and a follow-up week, and they reported daily self-compassion, perceived stress, and emotional exhaustion each evening. The analysis compared outcome levels and time trends across the three periods with hierarchical linear models and treated trait self-compassion as a person-level moderator. That design tests whether outcomes differ between the weeks of the program.

A phase comparison does not test whether a person's reported outcomes differ between the days they actually practice and the days they do not. Participants in the original study did not meditate every day (Li et al., 2021). Each morning of the intervention weeks they reported whether they had meditated in the previous 24 hours, and in the original report this item was used to describe compliance, not as a predictor. No analysis in the original report compares meditation days with nonmeditation days, models adherence or dose, or examines a lagged next-day association of meditation with exhaustion; its one-day lagged analyses relate self-compassion to next-day stress and exhaustion (Li et al., 2021).

There are substantive reasons to expect a same-day association. Guided self-compassion meditation focuses attention on kind and accepting self-talk, so the state the practice targets should be more available on a day of practice. A next-day association with emotional exhaustion is plausible for a related reason. Exhaustion persists from one day to the next, and self-compassion is linked to how people cope with stressful events (Allen & Leary, 2010, p. 107). Because this persistence is expected, any next-day test has to account for the previous day's exhaustion before attributing change to practice. The day-to-day persistence of affective states, often called inertia, is itself a feature of emotional functioning (Kuppens et al., 2010; Kuppens & Verduyn, 2017), so it is modeled rather than left in the residual.

The adherence question also raises a methodological problem. A diary lets each person serve as their own comparison. Centering a day-level predictor on each person's mean leaves no between-person variation in the predictor and yields a pure estimate of the within-person relationship (Enders & Tofighi, 2007, p. 128). The standard way to obtain that contrast is to subtract each person's observed mean of the predictor from the daily value. When a person contributes only a few days, however, the observed mean may be an unreliable measure of the person-level predictor, which biases the between-person estimate (Lüdtke et al., 2008). With a lagged outcome in the model, observed-mean centering also biases the autoregressive term (Hamaker & Grasman, 2015). Dynamic structural equation modeling addresses this by treating the person mean as a latent variable estimated jointly with the rest of the model (Asparouhov et al., 2018; Hamaker et al., 2018).

This problem is particularly important in the archival diary (Li et al., 2021). Meditation was common in the intervention period. The archival data file records 725 meditation reports, 147 nonmeditation reports, and 38 missing reports across the 910 intervention person-days, and

the original authors had already excluded participants who meditated fewer than seven times. Many participants therefore have few or no nonmeditation days, so their mean of the dichotomous meditation indicator is close to 1 and depends on a few informative days. A latent-centered model fits these data and makes the precision of the within-person estimate explicit.

In a secondary analysis of the 14 intervention days of the archival diary (Li et al., 2021), I estimate day-level associations between meditation and outcomes, using a latent-centered Bayesian model specified and time-stamped before any outcome was modeled. The research question was not addressed in the original study, and the prior use of the data is disclosed. The analysis adds a day-level adherence test to the original phase evaluation, and its estimates are within-person associations.

Hypothesis 1. Within persons, same-day self-compassion will be higher on days on which a person meditates than on days on which the same person does not.

Hypothesis 2. Within persons, meditating on a given day will be associated with lower emotional exhaustion on the following day, adjusting for exhaustion on the day of meditation.

Additional quantities are reported descriptively, including the between-person association of meditation propensity with outcomes, the day-to-day persistence (autoregression) of each outcome, and within-person variability. Perceived stress is a prespecified additional outcome. Its timing and the direction examined (lower stress with meditation) were set after the Hypothesis 1 and 2 results were known, and it was analyzed on the same day and on the next day; both results are reported.

Method

Transparency and Openness

This is a secondary analysis of an archival dataset that is openly available on the Open Science Framework at https://osf.io/pqf6v/ (Li, 2021; Li et al., 2021). The original report does not mention a trial registration. I wrote the analysis protocol, the Stan model, and the analysis code before fitting any model to outcome data, and I specified the analysis plan a priori, except the timing and sign of the perceived-stress analyses and the later removal of the ±0.2 *SD* band and of the prespecified posterior-probability decision rule. These exceptions and the changes between protocol versions are listed under Deviations From the Protocol. I inspected no individual data rows while writing the protocol, which was specified from the data file and the published article. The protocol, its version history, and all code are available in the project repository. I report how the analytic sample was determined, all exclusions (none were made, and no observations were dropped as outliers), all manipulations, and all measures, following the Journal Article Reporting Standards for quantitative research (JARS-Quant; Appelbaum et al., 2018). The archival data have no explicit license; they are used here for scholarly secondary analysis with citation.

Original Study, Participants, and Ethics

The original study recruited employees from various organizations by snowball sampling through social networks (Li et al., 2021); the report does not state how many employees were approached. The original report states neither the data-collection dates nor whether collection overlapped the COVID-19 pandemic; collection was completed before the article was received on August 1, 2020 (Li et al., 2021). At the end of the study, participants received the book *Wherever You Go, There You Are* (Li et al., 2021). Inclusion criteria were current employment,

age 18 years or older, and voluntary participation; exclusion criteria were lack of Internet access, inadequate proficiency in Chinese, previous participation in a meditation program longer than 8 weeks, and a diagnosed mental disorder or severe physical problem. Of 89 recruited employees, one did not complete baseline measures, nine dropped out, six reported meditating fewer than seven times during the intervention period, and eight received the wrong intervention materials, leaving 65 in the archival data (89 − 1 − 9 − 6 − 8 = 65). Of the nine dropouts, six left during the control week and three during the intervention period (Li et al., 2021, "Preliminary analyses" section); reasons for dropout are not reported. I report exact counts below 10 only for the exclusion flow and for Day 7 and Day 20 missingness of self-compassion, exhaustion, and perceived stress, because the source data are public and de-identified. All other counts below 10 are shown as <10 (minimum cell size of 10). Participant flow and missingness are reported following JARS-Quant (Appelbaum et al., 2018, pp. 7–8), the Consolidated Standards of Reporting Trials (CONSORT 2010; Schulz et al., 2010), and the Strengthening the Reporting of Observational Studies in Epidemiology statement (STROBE; von Elm et al., 2007). The final sample had a mean age of 35.65 years (SD = 6.19), and 63.1% were women. The original report does not give race or ethnicity. The study was approved by the Human Subject Review Board of the Department of Psychology of Sun Yat-sen University, and participants gave informed consent. The original study was supported by the National Natural Science Foundation of China (31700961), the Natural Science Foundation of Guangdong Province, China (2017A030310423), and the Fundamental Research Funds for the Central Universities, Sun Yat-sen University (20wkzd13; Li et al., 2021, Funding section). The original authors planned at least 74 participants on the basis of a power analysis for their phase contrasts; the archival sample fixed the present sample size, so I report precision in place of power (see Estimation and Precision).

Design and Procedure

The design was nonexperimental and longitudinal: a secondary analysis of the daily diary from a single-group intervention study with no control group. The diary began on the Monday after baseline measures and ran for 28 consecutive days in three periods: a 7-day control period with the diary only, a 14-day intervention period, and a 7-day follow-up period with the diary only. Each evening, participants completed the diary on a survey website. In the intervention period they first read materials introducing self-compassion and then followed a guided meditation, a body scan in the first intervention week and self-compassionate breathing in the second, both adapted from guided meditations by Kristin Neff. Each morning of the intervention period they reported whether they had performed the meditation during the past 24 hours (yes or no). The original report does not describe masking of participants or researchers, or any adverse events (Li et al., 2021).

Measures

Daily self-compassion was measured with the six-item Daily Self-Compassion Scale (Li et al., 2021), rated with reference to how participants treated themselves during difficult times that day, from 1 (*almost never*) to 5 (*almost always*). In the archival dataset, the composite is the sum of the six items, with Items 2, 4, and 6 reverse-scored, as the dataset's variable labels document (codebook; Li et al., 2021); reverse-scoring of negatively worded items follows the scoring of the Self-Compassion Scale (Neff, 2003, p. 234). The composite has a possible range of 6 to 30. *Daily emotional exhaustion* was measured with the five-item Emotional Exhaustion subscale of the Maslach Burnout Inventory–General Survey (Schaufeli et al., 1996, as cited in Li et al., 2021), rated for that day from 0 (*completely disagree*) to 6 (*completely agree*). The composite was computed in the archival dataset as the sum of the five items and has a possible

range of 0 to 30. The original report does not state which Chinese-language versions of the scales were administered (Li et al., 2021). The original composites were used as provided; how partially missing items entered them follows the original scoring, which is not documented. *Daily perceived stress* was measured with the four-item Perceived Stress Scale (Leung et al., 2010), rated from 0 (*never*) to 4 (*very often*); it is a prespecified additional outcome whose timing and sign were set post hoc. *Meditation* was the dichotomous morning report. Trait self-compassion was also measured at baseline with the 26-item Self-Compassion Scale (Li et al., 2021; Neff, 2003); it does not enter the present models.

Reliability was estimated separately within and between persons with two-level confirmatory factor analysis of self-compassion and exhaustion over the 14 intervention days (Days 7–20 of the diary). Omega at each level was computed by applying Equation 3 of Geldhof et al. (2014) to the level-specific estimates of a one-factor model per level, estimated in lavaan, Version 0.7-3 (Rosseel et al., 2026). Within-person omega was .60 for self-compassion and .89 for exhaustion; between-person omega was .75 and .97 (Table 1). Following Lai (2021), these between-person omegas describe latent person-level constructs and may overstate the reliability of observed person means of these scales, which were not analyzed. Over all 28 days, the self-compassion solution was inadmissible (see Supplemental Material), so omega is reported for the intervention days only. The original study reported within-person omegas of .886 for self-compassion, .849 for exhaustion, and .794 for perceived stress (Li et al., 2021), compared with the within-person omega of .60 for self-compassion here.

Analytic Sample and Alignment of Meditation Reports

The analysis used the 14 intervention days (Days 7–20; diary days are numbered 0–27) of all 65 participants (910 person-days); the control and follow-up weeks did not enter the models,

because meditation was prompted and reported only during the intervention. Because the meditation item was asked in the morning about the previous 24 hours, each morning report was assigned to the preceding diary day, so that meditation for day t was read from the report given on the morning of day $t + 1$. Under this rule the last intervention day would take its value from the first follow-up morning, when no report was collected, so that day's meditation status was treated as missing rather than as a "no." Reading the report as describing the same day was tested as a sensitivity analysis. No participant was excluded; participants with no nonmeditation days remained in the model, where they informed the person-level and autoregressive parameters, and their number is reported in Results. The original exclusion of participants who meditated fewer than seven times (Li et al., 2021) was already applied in the archival sample; a further refit applying that rule to the aligned data is sensitivity analysis S5. Analytic counts after alignment are reported in Results and Table 1.

Statistical Model

A two-level model of days within persons was fitted in Stan (Carpenter et al., 2017) via CmdStan 2.40.0, using the R packages cmdstanr 0.9.0 and posterior 1.6.1 in R 4.5.0 (versions as recorded in the project repository). The model was written for this study following dynamic structural equation modeling (Asparouhov et al., 2018; Hamaker et al., 2018). Each outcome was standardized over the observed values in each model's analysis period (Days 8–20 for Hypothesis 2 and the next-day perceived-stress model) so that coefficients are in units of the total (pooled) *SD* of the outcome. Slopes rescaled to within-person residual-*SD* units are given in the Table 2 note.

The model has two linked parts. In the first, each person's daily meditation report is modeled as a dichotomous (Bernoulli) outcome with a person-specific latent propensity, the

probability that this person meditates on a given day. In the second, the outcome on a given day is predicted from (a) the within-person meditation term, the difference between that day's report and the person's latent propensity; (b) the latent propensity itself, which represents between-person differences in how often people meditate; (c) a weekend indicator; (d) day in the program, centered at the middle of the study; and (e) the person's outcome on the previous day, expressed as a deviation from the person's own level, which captures day-to-day persistence with a common autoregressive coefficient. Each person has a random intercept and a random within-person meditation slope, with a correlation between them. For participants without nonmeditation days, the person-specific slope is informed only by the population distribution; they contribute to the intercept, propensity, and autoregressive parameters. The primary model retains these participants; no sensitivity fit excluding them was run. The fixed (average) within-person meditation slope is the estimand for both hypotheses.

For Hypothesis 1 the outcome is same-day self-compassion. For Hypothesis 2 the outcome is emotional exhaustion on day $t + 1$, the predictor is meditation on day t, and the autoregressive term is exhaustion on day t, so the meditation coefficient is adjusted for the previous day's exhaustion; on the first outcome day (Day 8), the first day of the analysis period, the autoregressive term is omitted, so that day is not adjusted for Day 7 exhaustion. Meditation reports were missing on 100 of the 910 person-days, 65 of them on Day 20 by design (Table 1). In the primary model, missing reports are flagged rather than coded as nonmeditation: Those days contribute to the outcome model but provide no within-person meditation information and are excluded from the meditation-propensity likelihood, which assumes that the reports are missing at random. The brms comparison model (S6) drops those days listwise. A sensitivity analysis for missing meditation reports was added after the data release (see Deviations From the

Protocol). Missing outcome values are estimated within the joint model, which is valid if they are missing at random given the variables in the model (Gelman et al., 2013, Section 8.2).

Prior Distributions and Estimation

Priors were weakly informative and centered on no association. The fixed slopes, including the within-person meditation slope, received normal priors centered at zero with a standard deviation of half an outcome standard deviation. The intercept, the autoregressive coefficient (bounded between −1 and 1), the standard deviations of the random intercept, random slope, and residual, and the mean log-odds of meditation received comparably weak normal or half-normal priors (on priors for standard deviations, see Gelman, 2006). The correlation between the random intercept and the random slope received a Lewandowski–Kurowicka–Joe (LKJ) prior that mildly favors small correlations. Exact prior specifications are in the protocol in the project repository, and prior and posterior densities of key parameters are in Supplemental Figure S2. Prior choice was informed by the When to Worry and How to Avoid the Misuse of Bayesian Statistics (WAMBS) checklist (Depaoli & van de Schoot, 2017). The posterior was sampled with the No-U-Turn sampler (Hoffman & Gelman, 2014) in four chains of 1,000 warm-up and 1,000 sampling iterations, with a target acceptance rate of .95 and maximum tree depth of 12. Convergence required a rank-normalized $\hat{R}$ below 1.01 (Vehtari et al., 2021, p. 671) and bulk and tail effective sample sizes above 400 (Vehtari et al., 2021, p. 672). Divergent transitions, which signal problems in the sampler's exploration of the posterior (Carpenter et al., 2017, Section 2.4; Gabry et al., 2019, Section 4), were also required to be absent, and any divergence was specified to trigger a refit with a target acceptance rate of .99 (Bürkner, 2017, p. 10). Posterior predictive checks compared the observed mean, standard deviation, and lag-1 autocorrelation of each outcome with those of data replicated from the posterior (Gabry et al.,

2019; Gelman et al., 2013, Section 6.3; Figure 1). A rerun of the analysis code reproduced the saved results exactly. An independent reimplementation of the Hypothesis 1 and 2 models agreed with the primary fit within Monte Carlo error (details in the project repository).

Estimation and Precision

Hypothesis 1 is primary; Hypothesis 2 is secondary and is reported on its own. The directional posterior probability in the prespecified direction, $P(b > 0)$ for Hypothesis 1 and $P(b < 0)$ for Hypothesis 2, is reported descriptively (Gelman et al., 2013, pp. 77, 95; Marsman & Wagenmakers, 2017, pp. 529–531). Here $P(\cdot)$ denotes a posterior probability; for example, $P(b < 0)$ is the posterior probability that the within-person slope is negative. The additional analyses are likewise reported descriptively (Rubin, 2021). For each estimate, the posterior median and the 95% equal-tailed credible interval (CrI) are also reported, following Bayesian reporting guidelines (Kruschke, 2021). Because nonmeditation days were few, precision is made explicit by reporting the posterior standard deviation and the width of the 95% CrI. I dropped the protocol's prespecified ±0.2 *SD* band because this report uses no effect-size cutoff; the band-based quantities remain in the released aggregates.

Sensitivity Analyses

Six sensitivity analyses (S1–S6) were specified in protocol Version 1 §9; Version 2 revised the S4 wording and implemented S6, and the protocol was locked before the first data release. Each was evaluated in the direction of the hypothesis it concerns, to check whether the results held under alternative assumptions (Cro et al., 2020, p. 2816; Little et al., 2012; Morris et al., 2014; Thabane et al., 2013); because S5 proved identical to the primary Hypothesis 1 model, five of the six are independent checks. S1 refits Hypothesis 1 with the prior *SD* of the slope coefficients (within-person slope, propensity, weekend, time, and lagged-covariate coefficients)

doubled (1.0) and halved (0.25). S2 reads each meditation report as describing the same day. S3 relaxes the missing-at-random assumption with a delta-adjusted pattern-mixture model in which missing self-compassion values are shifted 0.25 and 0.50 *SD* below the model prediction, following the general delta-adjustment approach to sensitivity analysis for missing data (Hayati Rezvan et al., 2018). S4 repeats Hypothesis 2 under the S1 priors, the S2 alignment, and missing exhaustion values shifted 0.50 *SD* above the prediction. S5 refits Hypothesis 1 after excluding participants with fewer than seven meditation days, the original rule (Li et al., 2021), applied to the aligned data. S6 replaces latent centering with conventional observed person-mean centering in brms 2.22.0 (Bürkner, 2017, 2018), regressing standardized self-compassion on the within-person and person-mean meditation terms, weekend, and centered time, with a random intercept and random meditation slope, first-order autoregressive residuals, and the same weakly informative priors on all four fixed coefficients. S6 is reported to show how much the centering choice matters; it never replaces the primary model.

Deviations From the Protocol

The protocol went through three time-stamped versions, all dated October 24, 2026, and was specified a priori (record in the project repository), except the timing and sign of the perceived-stress analyses, which were set after the Hypothesis 1 and 2 results were known. I also dropped the ±0.2 *SD* inconclusive band, prespecified in Version 1 and confirmed in Version 2, and the band-based quantities, because no effect-size cutoff is used; their values remain in the released aggregates. Version 2 raised the cutoff of the posterior-probability decision rule, and Version 3 corrected how the sign of each hypothesis is applied to that rule. I later dropped this prespecified posterior-probability decision rule, and the results are reported descriptively. The originally locked reliability computation omitted the factor variance term, giving within- and

between-person omegas of .986 and .912 for self-compassion and .967 and .973 for exhaustion; after the data release, I corrected the computation to implement Equation 3 of Geldhof et al. (2014) and report the corrected values. I added a sensitivity analysis for missing meditation reports after the data release and locked its analysis code before any run on the study data; its result will be reported in a revision. No refit at a higher acceptance rate was needed, because no fit had divergent transitions. For the brms comparison model (S6), the original fit saved only $\hat{R}$; its effective sample sizes were extracted post hoc from a seeded rerun of the locked analysis code, whose results were byte-identical to the original run. Two priors were set in the analysis code but not named in the protocol: a half-normal prior with scale 1 on the between-person standard deviation of meditation log-odds, and the same normal prior as the other slopes (SD = 0.5) on one further covariate coefficient, which has no data contribution and is sampled from its prior.

Results

Data Description and Compliance

After alignment, the 65 participants contributed 910 intervention person-days. Meditation was reported for 678 days, nonmeditation for 132 days, and the meditation report was missing for 100 days (Table 1). These counts differ from the same-row counts in the archival data file (725, 147, and 38) only because of the alignment: The three missing reports from the first intervention morning fall outside the analysis period, and the last intervention day, which has no following morning report, adds 65 missing values ($38 - 3 + 65 = 100$). Twelve participants had no nonmeditation day and were retained; all 65 had at least seven meditation days. Self-compassion and emotional exhaustion were each missing on 50 person-days. On the first intervention day (Day 7), self-compassion was missing for 2 of 65 participants and exhaustion

for 2 of 65, and the Day 8 morning report that supplies Day 7's meditation status was missing for fewer than 10 of 65 participants. On the last intervention day (Day 20), meditation status was missing for all 65 participants by construction, and self-compassion was missing for 3 of 65 participants and exhaustion for 3 of 65. Within- and between-person correlations are in Supplemental Table S3.

Convergence and Model Checks

The 14 Stan fits comprised the two primary models, all sensitivity models except S6 (S5 among them, although it duplicates the primary Hypothesis 1 model), and the two perceived-stress models (a prespecified additional outcome; timing and sign set post hoc). Across those 14 fits, computed over the key model parameters, every $\hat{R}$ was below the 1.01 criterion (maximum 1.01 when rounded to two decimals), the minimum bulk effective sample size was 602, and there were no divergent transitions, so the convergence criteria were met and no refit was needed (Supplemental Table S2, Supplemental Figure S1). For the 12 primary and sensitivity fits, the minimum tail effective sample size was 618; for the two perceived-stress fits, the minimum bulk effective sample size was 615. For the brms comparison model (S6), the maximum $\hat{R}$ saved with the original fit was 1.00; diagnostics extracted post hoc from a seeded rerun of the locked analysis code, whose results were byte-identical to the original run, gave a minimum bulk effective sample size of 1,310 (criterion 400). In posterior predictive checks with 400 replicated datasets per model, both primary models matched the mean and standard deviation of their outcome well (posterior predictive *p* values [Gelman et al., 1996], reported descriptively: .48 and .50 for Hypothesis 1, .52 and .55 for Hypothesis 2) but underestimated lag-1 autocorrelation. The replicated value reached the observed value in 0 of 400 replications: For Hypothesis 1 the

observed value was .88 versus a replicated median of .80, 95% interval [.76, .83], and for Hypothesis 2 it was .84 versus .73, 95% interval [.69, .77] (Figure 1).

Hypothesis 1: Meditation Days and Same-Day Self-Compassion

The within-person meditation slope for same-day self-compassion had a posterior median of 0.00 *SD*, posterior *SD* 0.05, 95% CrI [−0.09, 0.09]. The 95% CrI was centered on zero, and P($b > 0$) = .48 (Table 2, Figure 2).

Hypothesis 2: Meditation and Next-Day Emotional Exhaustion

Adjusting for same-day exhaustion, the within-person slope of meditation on next-day exhaustion had a posterior median of −0.03 *SD*, posterior *SD* 0.07, 95% CrI [−0.16, 0.11]. The 95% CrI included values of both signs, and P($b < 0$) = .65 (Table 2, Figure 2).

Sensitivity Analyses

For Hypothesis 1 (Table 3), doubling or halving the prior *SD* (S1) gave medians of 0.00 in both cases, P($b > 0$) = .48 and .49; reading meditation reports as same-day (S2) gave −0.02, 95% CrI [−0.12, 0.08], P($b > 0$) = .33; and shifting missing self-compassion values down by 0.25 or 0.50 *SD* (S3) gave 0.00 and −0.01, P($b > 0$) = .47 and .46. No participant met the S5 exclusion criterion, so S5 is identical to the primary Hypothesis 1 model and is not an independent check (Supplemental Material). With observed person-mean centering (S6), the median was 0.02, 95% CrI [−0.08, 0.11], P($b > 0$) = .64 (Supplemental Figure S3). For Hypothesis 2 (S4), the prior variations gave medians of −0.03 in both cases, P($b < 0$) = .66 and .67, and the +0.50 *SD* missing-data shift gave −0.03, P($b < 0$) = .65. Same-day alignment moved the Hypothesis 2 estimate furthest, to −0.07, 95% CrI [−0.18, 0.04], P($b < 0$) = .90.

Additional Analyses

Table 4 reports the additional quantities as posterior medians in total-*SD* units, with 95% CrIs. In the perceived-stress analysis (a prespecified additional outcome; timing and sign set post hoc), the within-person meditation slope for next-day perceived stress had a posterior median of 0.06 *SD*, posterior *SD* 0.08, 95% CrI [−0.09, 0.22], and P($b < 0$) = .21. For same-day perceived stress, the median was −0.10 *SD*, posterior *SD* 0.08, 95% CrI [−0.25, 0.06], and P($b < 0$) = .90. Both 95% CrIs included zero (Table 4; Supplemental Table S1).

Discussion

For same-day self-compassion (Hypothesis 1), the within-person meditation slope had a median of 0.00 *SD*, P($b > 0$) = .48, and the next-day exhaustion slope (Hypothesis 2) had P($b < 0$) = .65. Both within-person slopes had 95% CrIs spanning zero, [−0.09, 0.09] and [−0.16, 0.11]. These estimates come from a latent-centered model, chosen because nonmeditation days were few, and address whether, within the two weeks of an online self-compassion program, the days on which employees meditated differed from the days on which they did not.

Relation to the Original Phase Analysis

The original report describes changes in daily outcomes across the program phases (Li et al., 2021). Within the intervention period, there is little evidence that those changes were concentrated on the days on which participants actually meditated. The within-person difference in self-compassion between meditation and nonmeditation days (Hypothesis 1) was centered on zero, with an interval spanning both signs. One interpretation is that the changes during such a program reflect being in the program (reading the materials, expecting benefit, monitoring one's state, or the passage of time) rather than a given day's practice. Another is that any benefit of practice is cumulative over days and persists on days without practice, which a same-day

contrast cannot detect; the autoregressive term accounts for some day-to-day persistence but does not model cumulative dose. The single-group design cannot exclude time and monitoring effects (Li et al., 2021), but a day-level contrast within the intervention period holds them constant across meditation and nonmeditation days.

Methodological Implications

The latent-centered estimate (0.00) and the observed-centering estimate (0.02) differed by about 0.02 *SD*, and both 95% CrIs included zero, although 12 participants contributed no nonmeditation day. The largest shift came from the alignment of the morning report: Reading it as describing the same day moved the next-day exhaustion estimate (Hypothesis 2) from −0.03 to −0.07, so the mapping of an adherence item onto diary days should be stated and tested. Archival intervention diaries often contain adherence items that the original phase analysis did not use; such items can be analyzed as day-level predictors, provided the morning-report alignment, missing reports, and sparse nonadherence days are handled explicitly.

Limitations

Self-chosen timing means unmeasured demands could confound the association. The seven-meditation-day minimum left few nonmeditation days. Missing meditation reports were not imputed. Composites were computed in the archival dataset. Race and ethnicity were not reported in the original study or in the shared data. The model pools the two weeks' different meditations. Results from snowball-sampled Chinese employees may not generalize to other populations or formats. Underpredicted lag-1 autocorrelation (Figure 1) means slope intervals may be too narrow.

Future Directions

Because participants chose when to practice, designs that randomize practice days, such as microrandomized trials (Klasnja et al., 2015), would be needed to identify day-level effects of practice. The same-day perceived-stress slope, $P(b < 0) = .90$, is a candidate for a confirmatory test. Measuring daily demands would address time-varying confounding, and longer programs or samples without an adherence-based exclusion would yield more nonmeditation days per person and more precise within-person estimates. A prior predictive check and sensitivity analyses for the LKJ prior on the random-intercept–slope correlation and for the priors other than the fixed-slope priors (varied in S1 and S4) were not conducted.

Conclusion

In a secondary analysis of an archival 28-day diary, the days on which employees practiced self-compassion meditation did not differ clearly from their nonmeditation days in same-day self-compassion or next-day emotional exhaustion: Both within-person slopes had 95% CrIs spanning zero, and the Hypothesis 2 slope had a posterior median of −0.03 *SD*, with $P(b < 0) = .65$. Archival intervention diaries can thus answer adherence questions that their original analyses left open; here, the estimates gave no indication that the program's changes were concentrated on the days on which employees practiced.

References

Allen, A. B., & Leary, M. R. (2010). Self-compassion, stress, and coping. *Social and Personality Psychology Compass, 4*(2), 107–118. https://doi.org/10.1111/j.1751-9004.2009.00246.x

Appelbaum, M., Cooper, H., Kline, R. B., Mayo-Wilson, E., Nezu, A. M., & Rao, S. M. (2018). Journal article reporting standards for quantitative research in psychology: The APA Publications and Communications Board task force report. *American Psychologist, 73*(1), 3–25. https://doi.org/10.1037/amp0000191

Asparouhov, T., Hamaker, E. L., & Muthén, B. (2018). Dynamic structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal, 25*(3), 359–388. https://doi.org/10.1080/10705511.2017.1406803

Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software, 80*(1), 1–28. https://doi.org/10.18637/jss.v080.i01

Bürkner, P.-C. (2018). Advanced Bayesian multilevel modeling with the R package brms. *The R Journal, 10*(1), 395–411. https://doi.org/10.32614/RJ-2018-017

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software, 76*(1), 1–32. https://doi.org/10.18637/jss.v076.i01

Cro, S., Morris, T. P., Kenward, M. G., & Carpenter, J. R. (2020). Sensitivity analysis for clinical trials with missing continuous outcome data using controlled multiple imputation: A practical guide. *Statistics in Medicine, 39*(21), 2815–2842. https://doi.org/10.1002/sim.8569

Depaoli, S., & van de Schoot, R. (2017). Improving transparency and replication in Bayesian statistics: The WAMBS-Checklist. *Psychological Methods, 22*(2), 240–261. https://doi.org/10.1037/met0000065

Enders, C. K., & Tofighi, D. (2007). Centering predictor variables in cross-sectional multilevel models: A new look at an old issue. *Psychological Methods, 12*(2), 121–138. https://doi.org/10.1037/1082-989X.12.2.121

Gabry, J., Simpson, D., Vehtari, A., Betancourt, M., & Gelman, A. (2019). Visualization in Bayesian workflow. *Journal of the Royal Statistical Society Series A: Statistics in Society, 182*(2), 389–402. https://doi.org/10.1111/rssa.12378

Geldhof, G. J., Preacher, K. J., & Zyphur, M. J. (2014). Reliability estimation in a multilevel confirmatory factor analysis framework. *Psychological Methods, 19*(1), 72–91. https://doi.org/10.1037/a0032138

Gelman, A. (2006). Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper). *Bayesian Analysis, 1*(3), 515–534. https://doi.org/10.1214/06-BA117A

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian data analysis* (3rd ed.). CRC Press. https://doi.org/10.1201/b16018

Gelman, A., Meng, X.-L., & Stern, H. (1996). Posterior predictive assessment of model fitness via realized discrepancies. *Statistica Sinica, 6*(4), 733–760.

Hamaker, E. L., Asparouhov, T., Brose, A., Schmiedek, F., & Muthén, B. (2018). At the frontiers of modeling intensive longitudinal data: Dynamic structural equation models for the affective measurements from the COGITO study. *Multivariate Behavioral Research, 53*(6), 820–841. https://doi.org/10.1080/00273171.2018.1446819

Hamaker, E. L., & Grasman, R. P. P. P. (2015). To center or not to center? Investigating inertia with a multilevel autoregressive model. *Frontiers in Psychology, 5*, Article 1492. https://doi.org/10.3389/fpsyg.2014.01492

Hayati Rezvan, P., Lee, K. J., & Simpson, J. A. (2018). Sensitivity analysis within multiple imputation framework using delta-adjustment: Application to Longitudinal Study of Australian Children. *Longitudinal and Life Course Studies, 9*(3), 259–278. https://doi.org/10.14301/llcs.v9i3.503

Hoffman, M. D., & Gelman, A. (2014). The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research, 15*(47), 1593–1623.

Klasnja, P., Hekler, E. B., Shiffman, S., Boruvka, A., Almirall, D., Tewari, A., & Murphy, S. A. (2015). Microrandomized trials: An experimental design for developing just-in-time adaptive interventions. *Health Psychology, 34*(Suppl.), 1220–1228. https://doi.org/10.1037/hea0000305

Kruschke, J. K. (2021). Bayesian analysis reporting guidelines. *Nature Human Behaviour, 5*(10), 1282–1291. https://doi.org/10.1038/s41562-021-01177-7

Kuppens, P., Allen, N. B., & Sheeber, L. B. (2010). Emotional inertia and psychological maladjustment. *Psychological Science, 21*(7), 984–991. https://doi.org/10.1177/0956797610372634

Kuppens, P., & Verduyn, P. (2017). Emotion dynamics. *Current Opinion in Psychology, 17*, 22–26. https://doi.org/10.1016/j.copsyc.2017.06.004

Lai, M. H. C. (2021). Composite reliability of multilevel data: It's about observed scores and construct meanings. *Psychological Methods, 26*(1), 90–102. https://doi.org/10.1037/met0000287

Leung, D. Y. P., Lam, T.-H., & Chan, S. S. C. (2010). Three versions of Perceived Stress Scale: Validation in a sample of Chinese cardiac patients who smoke. *BMC Public Health, 10*(1), Article 513. https://doi.org/10.1186/1471-2458-10-513

Li, Y. (2021). *Daily interventions and assessments: The effect of online self-compassion meditation on psychological health* [Data set]. OSF. https://doi.org/10.17605/OSF.IO/PQF6V

Li, Y., Hu, Y., Yang, W., & Wang, Y. (2021). Daily interventions and assessments: The effect of online self-compassion meditation on psychological health. *Applied Psychology: Health and Well-Being, 13*(4), 906–921. https://doi.org/10.1111/aphw.12278

Little, R. J., D'Agostino, R., Cohen, M. L., Dickersin, K., Emerson, S. S., Farrar, J. T., Frangakis, C., Hogan, J. W., Molenberghs, G., Murphy, S. A., Neaton, J. D., Rotnitzky, A., Scharfstein, D., Shih, W. J., Siegel, J. P., & Stern, H. (2012). The prevention and treatment of missing data in clinical trials. *New England Journal of Medicine, 367*(14), 1355–1360. https://doi.org/10.1056/NEJMsr1203730

Lüdtke, O., Marsh, H. W., Robitzsch, A., Trautwein, U., Asparouhov, T., & Muthén, B. (2008). The multilevel latent covariate model: A new, more reliable approach to group-level effects in contextual studies. *Psychological Methods, 13*(3), 203–229. https://doi.org/10.1037/a0012869

Marsman, M., & Wagenmakers, E.-J. (2017). Three insights from a Bayesian interpretation of the one-sided *P* value. *Educational and Psychological Measurement, 77*(3), 529–539. https://doi.org/10.1177/0013164416669201

Morris, T. P., Kahan, B. C., & White, I. R. (2014). Choosing sensitivity analyses for randomised trials: Principles. *BMC Medical Research Methodology, 14*, Article 11. https://doi.org/10.1186/1471-2288-14-11

Neff, K. D. (2003). The development and validation of a scale to measure self-compassion. *Self and Identity, 2*(3), 223–250. https://doi.org/10.1080/15298860309027

Neff, K. D., & Germer, C. K. (2013). A pilot study and randomized controlled trial of the mindful self-compassion program. *Journal of Clinical Psychology, 69*(1), 28–44. https://doi.org/10.1002/jclp.21923

Rosseel, Y., Jorgensen, T. D., & De Wilde, L. (2026). *lavaan: Latent variable analysis* (Version 0.7-3) [Computer software]. https://lavaan.ugent.be

Rubin, M. (2021). When to adjust alpha during multiple testing: A consideration of disjunction, conjunction, and individual testing. *Synthese, 199*(3–4), 10969–11000. https://doi.org/10.1007/s11229-021-03276-4

Schulz, K. F., Altman, D. G., & Moher, D. (2010). CONSORT 2010 statement: Updated guidelines for reporting parallel group randomised trials. *BMJ, 340*, Article c332. https://doi.org/10.1136/bmj.c332

Thabane, L., Mbuagbaw, L., Zhang, S., Samaan, Z., Marcucci, M., Ye, C., Thabane, M., Giangregorio, L., Dennis, B., Kosa, D., Debono, V. B., Dillenburg, R., Fruci, V., Bawor, M., Lee, J., Wells, G., & Goldsmith, C. H. (2013). A tutorial on sensitivity analyses in

clinical trials: The what, why, when and how. *BMC Medical Research Methodology, 13*, Article 92. https://doi.org/10.1186/1471-2288-13-92

Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., & Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved R-hat for assessing convergence of MCMC (with discussion). *Bayesian Analysis, 16*(2), 667–718. https://doi.org/10.1214/20-BA1221

von Elm, E., Altman, D. G., Egger, M., Pocock, S. J., Gøtzsche, P. C., & Vandenbroucke, J. P. (2007). The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: Guidelines for reporting observational studies. *The Lancet, 370*(9596), 1453–1457. https://doi.org/10.1016/S0140-6736(07)61602-X

Table 1

Analytic Sample, Compliance, Descriptive Statistics, and Reliability (Intervention Period)

Quantity	Value
Persons / person-days	65 / 910
Meditation days	678
Nonmeditation days	132
Missing meditation reports	100
Persons with zero nonmeditation days	12
Persons with < 7 meditation days	0
SC / EE missing days	50 / 50
PS missing days, same-day / next-day	50 / 48
SC *M* (*SD*), ICC	20.82 (3.49), .80
EE *M* (*SD*), ICC	10.69 (5.34), .72
PS *M* (*SD*), ICC	6.09 (2.33), .54
ω within: SC / EE	.60 / .89
ω between: SC / EE	.75 / .97

Note. Counts after previous-day alignment (meditation on day *t* read from the morning report of day *t* + 1; the last intervention day set to missing). Descriptive statistics are for Days 7–20 on the raw composite scale; the ICC is the between-person share of total variance (intercept-only model). SC = daily self-compassion; EE = emotional exhaustion; PS = perceived stress; ICC = intraclass correlation; ω = level-specific omega.

Table 2

Within-Person Meditation Slopes for Same-Day Self-Compassion (H1) and Next-Day Emotional Exhaustion (H2)

Parameter	Median	Post. *SD*	95% CrI	P(direction)	$\hat{R}_{max}$	ESS_{min}
H1: b_{within}, SC (t)	0.00	0.05	[−0.09, 0.09]	.48	1.00	1,209/2,210
H2: b_{within}, EE ($t + 1$)	−0.03	0.07	[−0.16, 0.11]	.65	1.00	680/780

Note. Slopes are in units of the total (pooled) *SD* of the outcome. Credible intervals (CrIs) are equal-tailed quantile intervals. Rescaled by the model's within-person residual *SD*, the slopes (posterior median [95% CrI]) are −0.01 [−0.21, 0.21] for H1 and −0.06 [−0.33, 0.23] for H2. P(direction) = posterior probability that b lies in the prespecified direction (> 0 for H1, < 0 for H2). Results are reported descriptively. Across all 14 Stan fits, including the two perceived-stress fits (prespecified additional outcome; timing and sign set post hoc), and computed over the key model parameters, every $\hat{R}$ was below 1.01 (maximum 1.01 when rounded to two decimals), the minimum bulk ESS was 602 (615 in the perceived-stress fits), and there were no divergent transitions. Autoregression φ estimates are in Table 4. H1 = Hypothesis 1; H2 = Hypothesis 2; SC = daily self-compassion; EE = emotional exhaustion; b_{within} = within-person slope; Post. *SD* = posterior standard deviation; ESS = effective sample size; $\hat{R}_{max}$ = maximum rank-normalized $\hat{R}$; ESS_{min} = minimum ESS, bulk/tail.

Table 3

Sensitivity Analyses

Model	Hypothesis	Median	95% CrI	P(direction)	$\hat{R}_{max}$
S1: prior *SD* 1.0	H1	0.00	[−0.10, 0.09]	.48	1.00
S1: prior *SD* 0.25	H1	0.00	[−0.10, 0.09]	.49	1.00
S2: same-day alignment	H1	−0.02	[−0.12, 0.08]	.33	1.00
S3: SC δ = −0.25 *SD*	H1	0.00	[−0.10, 0.09]	.47	1.00
S3: SC δ = −0.50 *SD*	H1	−0.01	[−0.10, 0.09]	.46	1.00
S4: prior *SD* 1.0	H2	−0.03	[−0.17, 0.11]	.66	1.01
S4: prior *SD* 0.25	H2	−0.03	[−0.16, 0.10]	.67	1.00
S4: same-day alignment	H2	−0.07	[−0.18, 0.04]	.90	1.00
S4: EE δ = +0.50 *SD*	H2	−0.03	[−0.16, 0.11]	.65	1.01
S6: observed centering (brms)	H1	0.02	[−0.08, 0.11]	.64	1.00

Note. S5 is not listed: No participant met its exclusion criterion (fewer than seven meditation days), so it is identical to the primary H1 model (Table 2) rather than an independent check. S6 is a comparison model and never replaces the primary latent-centered model; $\hat{R}$ for S6 was saved with the original fit; its effective sample size (minimum bulk 1,310, criterion 400) was extracted post hoc from a seeded rerun of the locked analysis code, whose results were byte-identical to the original run. $\hat{R}$ values are rounded to two decimals; all were below 1.01 before rounding. P(direction) = posterior probability that *b* lies in the prespecified direction (> 0 for H1, < 0 for H2). H1 = Hypothesis 1; H2 = Hypothesis 2; SC = daily self-compassion; EE = emotional exhaustion; CrI = credible interval; δ = pattern-mixture shift applied to missing outcome values.

Table 4

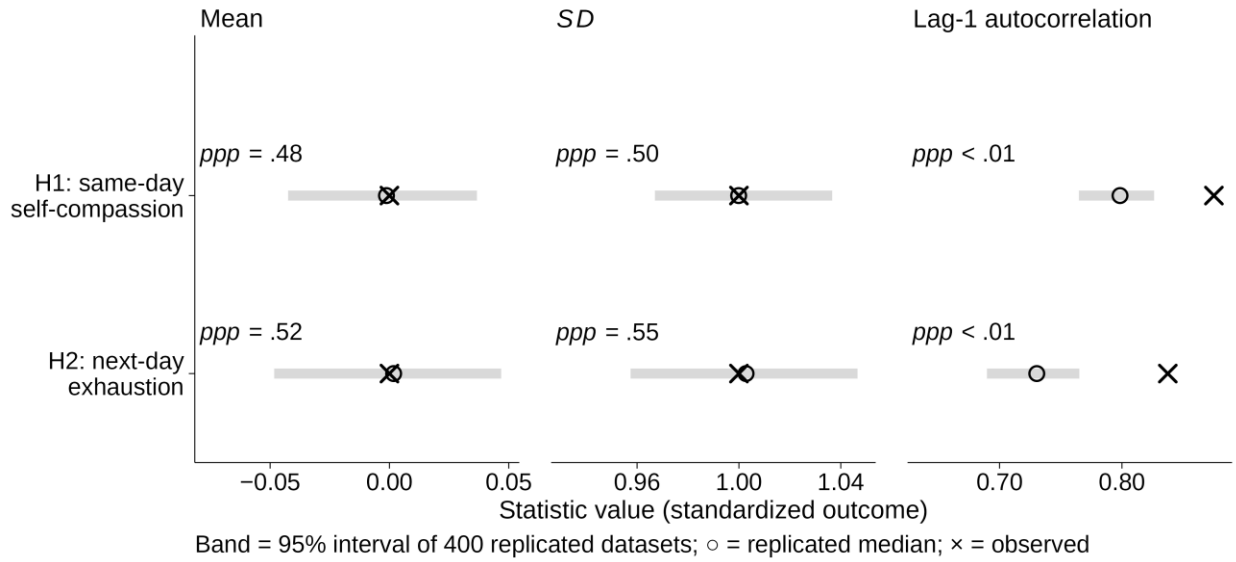
Additional Posterior Quantities

Quantity	SC (95% CrI)	EE (95% CrI)	PS next day (95% CrI)
Between-person coefficient *g*	0.15 [−0.38, 0.73]	−0.06 [−0.50, 0.39]	0.01 [−0.86, 0.93]
Autoregression φ	.37 [.29, .46]	.48 [.39, .56]	.17 [.08, .26]
Residual *SD* σ	0.43 [0.41, 0.45]	0.48 [0.45, 0.51]	0.65 [0.62, 0.69]
Within-person slope	(Table 2)	(Table 2)	0.06 [−0.09, 0.22]

Note. Values are posterior medians with 95% credible intervals (CrI). Reported descriptively (Rubin, 2021). All quantities are in units of the total (pooled) *SD* of the outcome; the between-person coefficient *g* is the outcome difference per unit of a person's latent meditation probability (0 = never, 1 = always). SC = daily self-compassion; EE = emotional exhaustion; PS = perceived stress, a prespecified additional outcome, analyzed on the day after the meditation report; the timing and sign of the PS analysis were set post hoc, after the Hypothesis 1 and 2 results were known. Same-day PS estimates are in Supplemental Table S1.

Figure 1

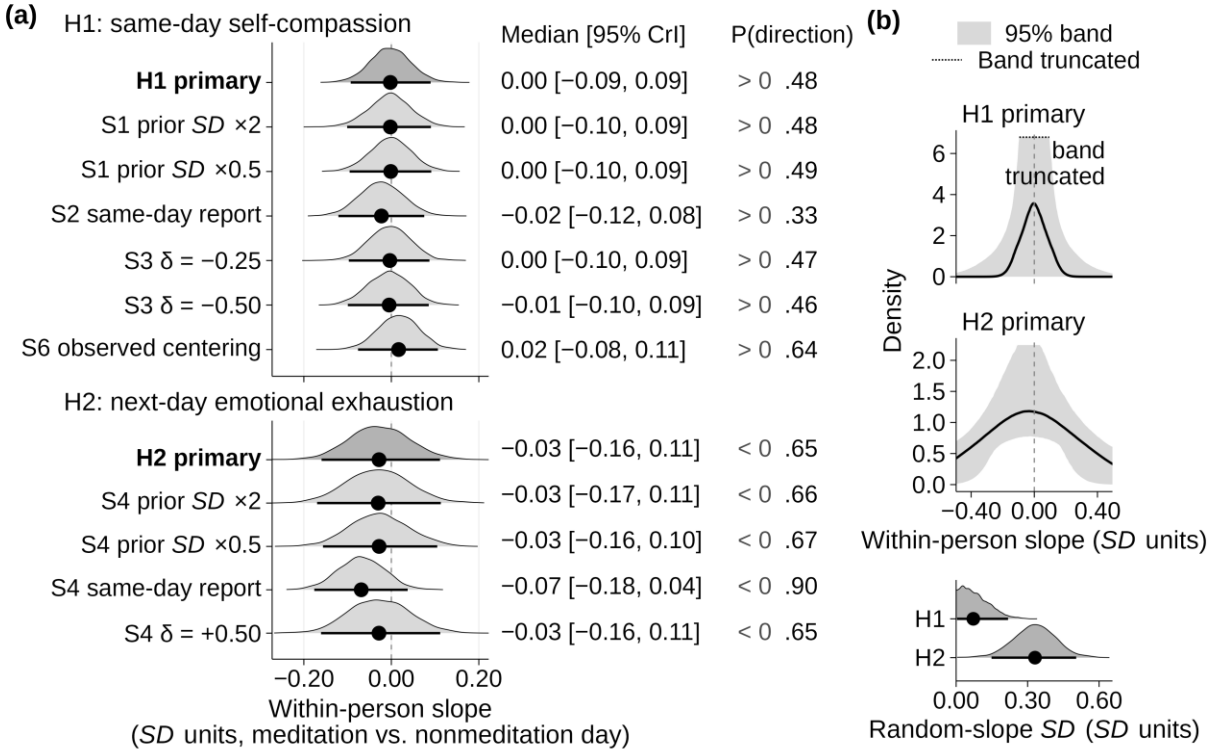
Posterior Predictive Checks of the Mean, Standard Deviation, and Lag-1 Autocorrelation



Note. $N = 65$ participants (910 person-days); 400 datasets were replicated from each model's posterior. Each row compares the observed statistic of the standardized outcome (×) with the median (open circle) and 95% equal-tailed interval (gray bar) of the replicated statistics. *ppp* = posterior predictive *p* value, the proportion of replicated statistics greater than or equal to the observed one. H1 = Hypothesis 1 model (same-day self-compassion); H2 = Hypothesis 2 model (next-day emotional exhaustion); both are two-level Bayesian latent-centered models. Mean and *SD* are reproduced well (*ppp* = .48–.55); lag-1 autocorrelation is underpredicted. *SD* = standard deviation.

Figure 2

Posterior Distributions of the Within-Person Meditation Slope Across Models and Implied Variation in Person-Specific Slopes



Note. $N = 65$ participants; *SD* units. (a) Within-person slope b (meditation vs. nonmeditation day): medians (points), 95% CrIs (lines), zero (dashed). P(direction) = posterior probability of the prespecified direction. S1 = prior *SD* ×2 or ×0.5; S2 = same-day alignment; S3 = missing values shifted δ = −0.25 or −0.50 *SD*; S4 = H2 checks; S6 = observed person-mean centering. (b) Person-specific slopes, $b_i \sim N(b_0$, random-slope *SD*): median (solid), pointwise 95% band; inset = random-slope *SD*. Band truncated = band capped at 1.9 × peak median density (dotted line, H1 cap). H1/H2 = Hypothesis 1/2; *SD* = standard deviation; CrI = credible interval.

Does Daily Meditation Increase Self-Compassion? A Bayesian Multilevel Diary Study

Supplemental Material

Sensitivity Analysis S5

No participant met the S5 exclusion rule (all 65 participants had at least seven meditation days in the intervention period, Days 7–20; with previous-day alignment, Day 20 is missing, so the count covers Days 7–19), so S5 is identical to the Hypothesis 1 model and is not reported as a separate result or counted as an independent check.

Sensitivity Analysis S6

The S6 comparison model converged. The original fit saved only $\hat{R}$ (maximum 1.00); diagnostics extracted post hoc from a seeded rerun of the locked analysis code, whose results were byte-identical to the original run, gave, for the key model parameters, $\hat{R}$ of 1.00 and minimum bulk and tail effective sample sizes of 1,310 and 1,899 (criterion 400; Table S2).

Data Structure, Missing Data, and Source Notes

Missing meditation reports were not imputed, which assumes they were missing at random with respect to daily self-compassion. The model uses a complete person-day grid (65 × 14 = 910 person-days, 845 one-day lag pairs), so every lag is exactly 1 day. On the 100 days with a missing meditation report in the Hypothesis 1 model, the report was flagged as missing rather than coded as nonmeditation and excluded from the meditation-propensity likelihood, and those days added no within-person meditation information, at some cost in precision. The 50 missing self-compassion values were estimated within the model, so 1-day lags that span missing outcome days depend on that estimation; the delta-shift sensitivity analyses (S3 and S4) test the missing-data assumption for self-compassion and exhaustion values only. Dropping missing days instead would make 4.28% of intervals (34 of 795) longer than 1 day. Four further limitations

concern the source data and code. First, the self-compassion and exhaustion composites were computed in the archival dataset, and how partially missing items were handled follows the original scoring, which is not documented. Second, the Day 8 outcome in the Hypothesis 2 model is not adjusted for Day 7 exhaustion, because the autoregressive term is omitted on the first day of each person's model window (Days 8–20). Third, some priors were taken from the analysis code rather than from the time-stamped protocol. Fourth, the source does not document the label for the time of each report or the coding of the weekday/weekend indicator. Meditation was reported the next morning; the same-day alignment is tested as a sensitivity analysis (S2 and S4). Because the original sample retained only participants who meditated at least seven times, the results may not generalize to less adherent employees.

Additional Perceived-Stress Analysis, Same Day

Perceived stress was also analyzed on the same day as the meditation report (additional, prespecified outcome; timing and sign set after the Hypothesis 1 and 2 results were known). The within-person meditation slope had a posterior median of −0.10 *SD*, posterior *SD* 0.08, 95% CrI [−0.25, 0.06], and P($b < 0$) = .90 (Table S1).

Table S1

Same-Day Perceived Stress Model (Additional Analysis)

Quantity	PS, same day (95% CrI)
Between-person coefficient *g*	−0.11 [−0.97, 0.76]
Autoregression φ	.18 [.10, .26]
Residual *SD* σ	0.66 [0.63, 0.69]
Within-person slope	−0.10 [−0.25, 0.06]

Note. Posterior medians with 95% credible intervals (CrI), in units of the total (pooled) *SD* of perceived stress (PS). Reported descriptively.

Multilevel Reliability

Over all 28 days, the two-level solution for self-compassion was inadmissible because the between-level residual variance of Item 5 was not positive, so it is not reported, and the intervention-period estimates (Days 7–20) are primary. Item 1 of the self-compassion scale had no within-person variance for some participants.

Convergence Diagnostics

Table S2 reports convergence diagnostics for the two primary models and the S6 comparison model, and Figure S1 shows the corresponding rank plots. Figure S2 shows prior and posterior distributions of key parameters, and Figure S3 shows within- and between-person meditation effects under latent and observed centering.

Table S2

Convergence Diagnostics for the Primary Models and S6

Model	$\hat{R}_{max}$	Min bulk ESS	Min tail ESS	Divergences	Max-treedepth hits	MCSE
H1 (primary)	1.00	1,209	2,210	0	not available	not available
H2 (secondary)	1.00	680	780	0	not available	not available
S6 (comparison, brms)	1.00	1,310	1,899	0	0	not available

Note. For all three models, values are taken over the key model parameters (within-person slope; between-person coefficient [propensity *g* for H1 and H2; observed person mean for S6]; autoregression φ; random-slope *SD*; residual *SD*). For the H1 within-person slope alone, $\hat{R}$ was 1.00, bulk ESS 6,426, and tail ESS 3,295. $\hat{R}$ values are rounded to two decimals. Maximum-treedepth hits were not saved for H1 and H2. S6 diagnostics were extracted post hoc from a seeded rerun of the locked analysis code, whose results were byte-identical to the original run. H1 = Hypothesis 1; H2 = Hypothesis 2; ESS = effective sample size; MCSE = Monte Carlo standard error.

Table S3

Within- and Between-Person Correlations

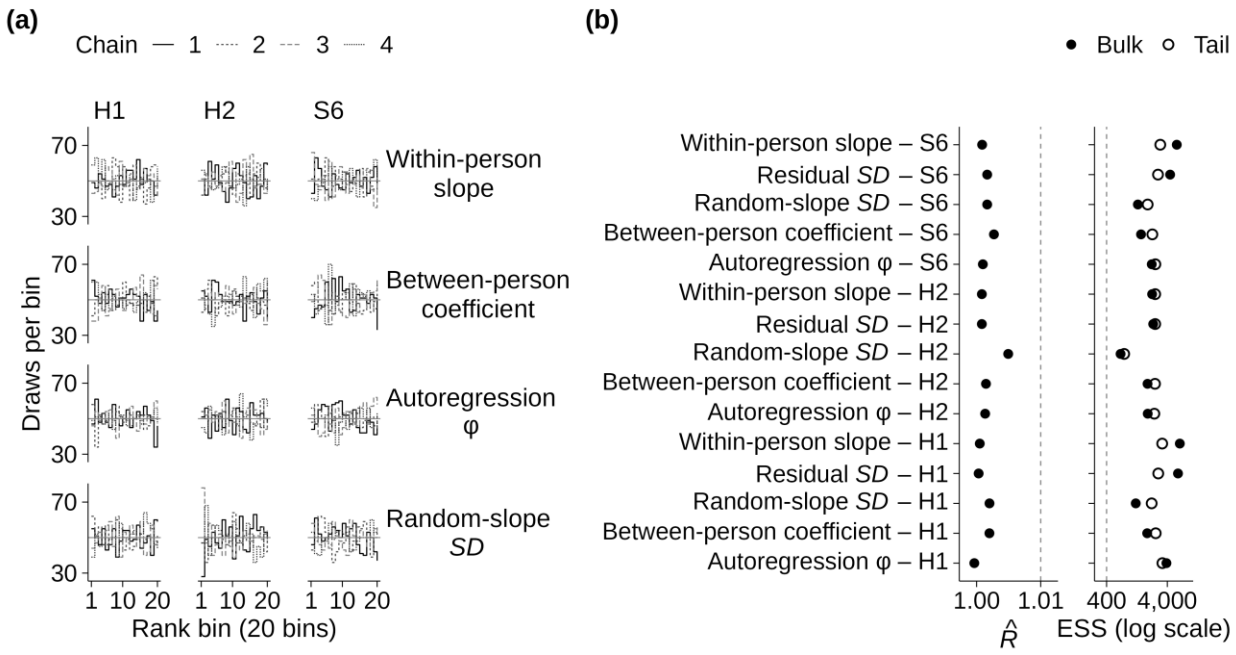
Variable	1	2	3	4
1. Self-compassion	—	−.32	−.28	[a]
2. Emotional exhaustion	−.48	—	.48	[a]
3. Perceived stress	−.54	.72	—	[a]
4. Meditation	.00	.12	−.09	—

Note. Pearson correlations for Days 7–20. Values above the diagonal are pooled within-person correlations of person-mean-centered daily values (n = 860 person-days); values below the diagonal are between-person correlations of person means (n = 65). Meditation is the same-day report (coded 0/1), not the previous-day alignment used in the models; its between-person value is the proportion of days with meditation. Correlations are descriptive and uncorrected. Dashes on the diagonal mark self-correlations, which are not shown. The association of meditation with self-compassion (Hypothesis 1) is tested in the Bayesian multilevel model.

Cells marked [a] are suppressed under the minimum-cell-size rule (any count, or count derivable from public totals, below 10).

Figure S1

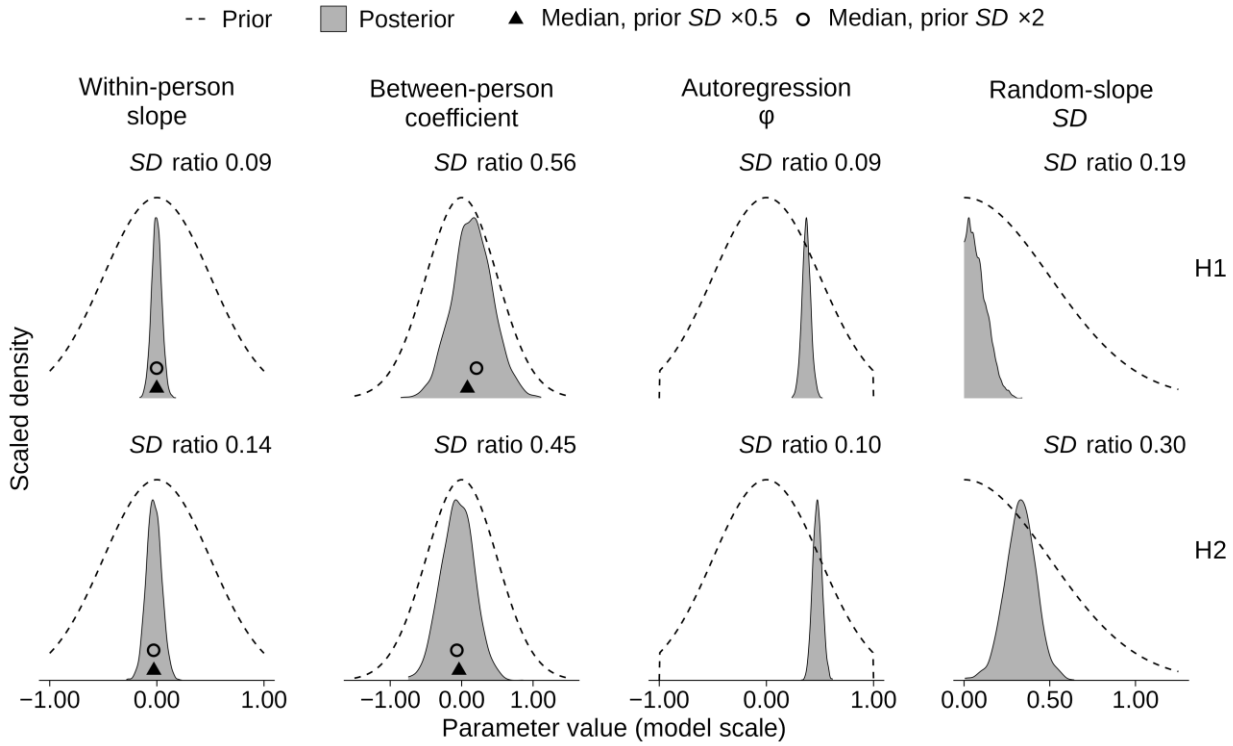
Rank Plots and Convergence Diagnostics for the Primary, Secondary, and Observed-Centering Models



Note. N = 65 participants (910 person-days). (a) Rank plots: post-warm-up draws of four chains ranked jointly, histogrammed per chain (line type); flat, overlapping lines near the dotted uniform expectation indicate good mixing. (b) Rank-normalized $\hat{R}$ and bulk (filled) and tail (open) effective sample size (ESS) for each model's key parameters; dashed lines = prespecified criteria ($\hat{R} < 1.01$, ESS > 400). Each model: 4 chains × 2,000 iterations (1,000 warm-up). $\hat{R}$ = potential scale reduction factor; Within-person slope = meditation-day minus nonmeditation-day difference; Between-person coefficient = difference per unit of meditation propensity (latent *g* for H1 and H2; observed person mean for S6); Autoregression φ = coefficient of the previous day's deviation from the person's mean; Random-slope *SD* = *SD* of person-specific slopes; Residual *SD* = *SD* of day-level residuals (panel b only). H1/H2 = Hypothesis 1/2; S6 = observed person-mean centering (brms). Exact values are in Table S2.

Figure S2

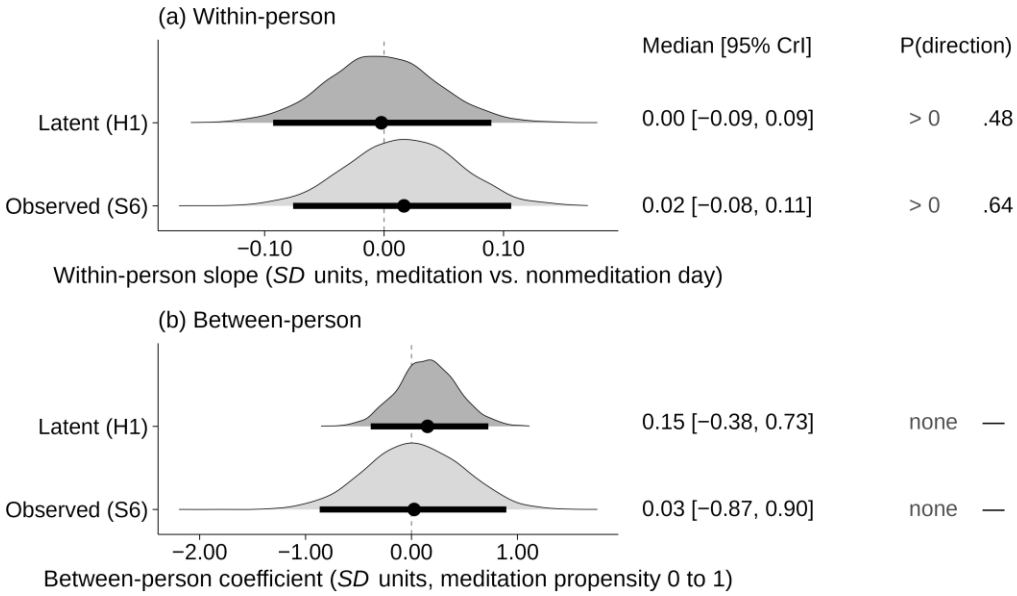
Prior and Posterior Distributions of Key Parameters



Note. $N = 65$ participants (910 person-days). Dashed lines are prior densities and gray areas are posterior densities, scaled to equal height, for the primary (H1 = Hypothesis 1, same-day self-compassion) and secondary (H2 = Hypothesis 2, next-day emotional exhaustion) models. Priors: normal(0, 0.5) for the within-person slope and the between-person coefficient; normal(0, 0.5) truncated to (−1, 1) for autoregression φ; half-normal(0, 0.5) for the random-slope *SD*. A filled triangle marks the posterior median when the prior *SD* of the slope coefficients was halved (×0.5, 0.25) and an open circle when it was doubled (×2, 1.0) (S1, S4). Median markers are shown for the within-person slope and between-person coefficient only. Slopes, coefficients, and *SD*s are in *SD* units; φ is unitless. *SD* ratio = posterior *SD* / prior *SD*; *b* = within-person slope; φ = autoregression coefficient; *g* = between-person coefficient (propensity).

Figure S3

Within- and Between-Person Meditation Effects Under Latent and Observed Centering



Note. $N = 65$ participants (910 person-days); 12 had no nonmeditation day. Posterior densities, medians (points), and 95% equal-tailed credible intervals (lines) of same-day self-compassion effects, in *SD* units. Within-person slope (*SD* units, meditation vs. nonmeditation day) = the difference in self-compassion between meditation and nonmeditation days (0/1 day indicator centered on the person's propensity). Between-person coefficient (*SD* units, meditation propensity 0 to 1) = the same parameter as *g* in Table 4 of the main text: difference in self-compassion per unit of a person's latent meditation probability (H1) or observed meditation proportion (S6) (0 = never, 1 = always); Latent = primary model (H1 = Hypothesis 1); Observed = observed-centering model (S6, brms). (a) = within-person slopes; (b) = between-person coefficients. P(direction) = posterior probability that *b* lies in the prespecified direction (> 0 for H1, < 0 for H2); “none” and the dash (between-person rows) = no prespecified direction. CrI = credible interval. The dashed line marks zero.